

Regime-Dependent Performance of ARIMA and Modern Forecasting Methods: An Empirical Benchmark on Small-Scale Retail Demand Data

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Abstract—Choosing between classical statistical models and modern machine learning for retail demand forecasting is a common but underspecified decision in practice. This paper presents a systematic empirical benchmark comparing six methods — ARIMA, SARIMA, Prophet, XGBoost, LSTM, and an ARIMA+XGBoost decomposition (Hybrid) — across five dataset sources spanning three countries and four demand regimes: (1) four real product-category series from a D-Mart retail store (India, daily, $n=181$), (2) five real product series from UCI Online Retail (UK, weekly, $n=53$), (3) 24 real series from the M4 Micro Monthly benchmark (international, $n=68-197$), (4) a weekly Walmart-calibrated store series ($n=143$), and (5) a daily intermittent demand series calibrated to M5 statistics ($n=365$, zero-fraction 32.9%). All models use fixed practitioner-default hyperparameters to reflect realistic deployment; all pairwise comparisons are tested with the Diebold-Mariano (DM) test.

Within the scope of these six datasets — short series ($n \leq 200$), daily or weekly retail demand, limited seasonality — we observe consistent regime-dependent performance. On the real D-Mart data ($CV \approx 0.021$, $|AC(1)| < 0.21$), classical ARIMA matches or outperforms all complex methods, which overfit the limited temporal signal; this holds under 5-fold walk-forward cross-validation and across 17 of 20 individually tested SKUs. On the Walmart data ($CV \approx 0.31$), the Hybrid reduces RMSE by 49.6% over ARIMA (83.15 vs 165.23). On M5 (zero-fraction 32.9%), all methods converge.

Based on these observations, we propose CV as a *preliminary heuristic* for model selection: $CV < 0.03$ consistently favored ARIMA in our study; $CV \approx 0.08-0.31$ favored the Hybrid on structured data. We explicitly note this heuristic is derived from six datasets under constrained conditions and should *not* be treated as a universal rule. A sensitivity analysis and synthetic experiment provide supporting evidence while quantifying the heuristic’s uncertainty. All results are statistically significant ($p < 0.001$, DM test). Code and data: <https://github.com/Aarav500/retail-forecasting-benchmark>.

Index Terms—Demand forecasting, ARIMA, XGBoost, LSTM, Prophet, hybrid models, Diebold-Mariano test, intermittent demand, retail analytics, time series benchmarking, regime characterization

I. INTRODUCTION

Accurate demand forecasting is foundational to retail operations, directly determining inventory decisions, promotional planning, and supply chain coordination [1], [17]. Despite decades of academic work and increasing deployment of machine learning in industry, a critical practical question remains inadequately studied: *under what conditions do mod-*

ern machine learning methods actually outperform classical statistical models on small-scale retail data?

Existing large-scale benchmarks — M4 [2] and M5 [3] — evaluate thousands of series where neural methods have demonstrated competitive performance. Recent Transformer-based approaches [20], [21] have achieved state-of-the-art results on these benchmarks. However, the vast majority of real-world retail deployments involve far smaller datasets: a single store’s six months of daily records, a regional chain’s two years of weekly sales. This regime — short observation windows, limited temporal autocorrelation, heterogeneous product categories — is underrepresented in the literature and presents distinct modeling challenges. In this regime, model complexity becomes a liability rather than an asset: overfitting to noise degrades performance relative to simple baselines [4].

This paper makes five contributions:

- 1) **Multi-source real-world benchmark:** We evaluate across five dataset sources spanning three countries — D-Mart (India, daily), UCI Online Retail (UK, weekly), M4 Micro (international, monthly), Walmart-calibrated (US, weekly), and M5-calibrated (intermittent) — covering 34 individual time series across four distinct demand regimes.
- 2) **Standardized six-model comparison:** ARIMA, SARIMA, Prophet, XGBoost, LSTM, and Hybrid are evaluated with fixed practitioner-default hyperparameters and strict temporal train/test splits across six datasets.
- 3) **ARIMA+XGBoost Hybrid evaluation:** We evaluate the established ARIMA+ML decomposition [10] using XGBoost as the residual corrector, providing the first DM-tested evaluation of this approach on real Indian retail data.
- 4) **Statistical significance via DM testing:** All pairwise comparisons use the Diebold-Mariano test [9], which is frequently omitted from applied forecasting benchmarks [5].
- 5) **Heuristic CV criterion with sensitivity analysis:** We propose CV as a preliminary model-selection indicator and support it with a sensitivity analysis and synthetic experiment — while explicitly scoping it as a heuristic derived from limited data, not a validated universal rule.

II. BACKGROUND AND RELATED WORK

A. Classical Time Series Methods

Box and Jenkins [1] established the ARIMA framework through autoregressive modeling, differencing, and moving average components. Seasonal extensions (SARIMA) address periodic retail cycles [5]. Despite their age, these models remain competitive in small-data regimes where sample complexity limits learning-based methods [4]. The M4 competition [2] — covering 100,000 heterogeneous time series — found that simple methods outperform complex models on short series, a key motivation for our low-SNR regime analysis.

B. Machine Learning for Forecasting

XGBoost [6] with lag feature engineering has demonstrated strong performance on tabular time series and won multiple M5 competition tracks [3]. Prophet [7] provides an additive decomposition model designed for business time series with irregular seasonality. LSTM networks [8] capture long-range temporal dependencies but require substantial data for reliable training [12].

Transformer-based architectures have recently achieved state-of-the-art results on large-scale benchmarks. Wen et al. [20] survey Transformer variants for time series, finding consistent gains on datasets with thousands of training points. Ekambaram et al. [21] introduce Tiny Time Mixers achieving strong zero-shot performance. Critically, these results are obtained on datasets far larger than the small-scale retail regime we study; as [4] demonstrate, gains from complex models diminish sharply as series length decreases.

A 2024 retail-specific benchmark by Kaur et al. [22] evaluates Transformer, Informer, PatchTST, and TFT against ARIMA on the full M5 dataset (30,490 series), reporting 26–29% MASE improvements for Transformers. This is consistent with our findings on large structured data (Walmart) but does not address the short-series, single-category regime central to our contribution.

C. Hybrid Methods

Zhang [10] proposed the foundational ARIMA+neural network hybrid, arguing that ARIMA captures linear autocorrelation while neural components model nonlinear residuals. Khashei and Bijari [11] extended this framework with ARIMA+MLP hybrids, demonstrating improvements on benchmark datasets. A 2024 study [24] applied hybrid CNN-LSTM approaches to multi-channel retail sales, finding improvements over pure neural approaches. Our Hybrid (ARIMA+XGBoost) extends this literature with a more computationally efficient tree-based residual corrector and provides the first systematic evaluation under DM statistical testing on real Indian retail data.

D. Benchmarking, Regime Awareness, and Evaluation

Makridakis et al. [2], [3] established the gold standard for forecasting benchmarks. Cerqueira et al. [19] studied cross-validation strategies for time series, finding that improper

evaluation protocols can reverse benchmark conclusions — motivating our strict temporal split protocol. Diebold and Mariano [9] provide the standard framework for testing predictive accuracy differences; its frequent omission from applied benchmarks is a known methodological gap [5].

A key insight from recent work is that model selection must be *regime-aware*. Nasser et al. [23] found tree-based methods (ETR) outperform LSTM on a 5.2-million-record retail dataset with perishable products. Conversely, de Castro Moraes et al. [24] found hybrid CNN-LSTM models superior for multi-channel fashion retail with sufficient data. Neither study provides a unifying characterization of when each approach dominates, which our CV-based criterion addresses.

III. DATASET DESCRIPTION AND CHARACTERIZATION

A. Real D-Mart Dataset

The primary dataset consists of retail sales records from a D-Mart store (Pune, India), covering January 1 to June 30, 2023. The full dataset contains 144,800 records spanning 800 SKUs across four product categories. Each record includes: Date, Product_ID, Inventory_Level, Sales, Lead_Time, Price, Promotion_flag, Season, External_Factors, Supplier_Reliability, Stock_Replenishment, Demand_Forecast, Order_Frequency, and Product_Category.

For time series forecasting, we aggregate daily sales by product category, yielding four series of $n=181$ observations each. Table I summarizes key properties. All four series are stationary by ADF test ($p < 0.001$), exhibit near-zero lag-1 autocorrelation ($|AC(1)| < 0.21$), and have $CV \approx 0.021$ — indicating weak temporal signal relative to noise.

TABLE I
DATASET OVERVIEW

Dataset	Source	n	Freq	Mean CV	Mean AC(1)	Real?	Regime
D-Mart Food	India, single store	181	Daily	0.021	−0.036	Yes	Low-SNR
D-Mart Electronics	India, single store	181	Daily	0.021	−0.166	Yes	Low-SNR
D-Mart Clothing	India, single store	181	Daily	0.022	−0.091	Yes	Low-SNR
D-Mart Furniture	India, single store	181	Daily	0.022	−0.209	Yes	Low-SNR
UCI Retail (5 series)	UK, online retail	53	Weekly	0.45	0.51	Yes	High-var
Walmart	US, calibrated [16]	143	Weekly	0.31	0.71	Calibrated	High-var
M4 Micro (24 series)	International, M4 [2]	68–197	Monthly	0.20	0.88	Yes	Structured
M5 Intermittent	Calibrated [3]	365	Daily	N/A	0.03	Calibrated	Intermittent

The D-Mart dataset also contains exogenous variables (Promotion, Price, External_Factors). Preliminary experiments with ARIMAX — incorporating these covariates — yielded RMSE of 450.98 on Food vs. 255.76 for plain ARIMA, indicating the covariates add noise rather than signal at this aggregation level. This is consistent with the low correlation between Promotion and Sales ($r = -0.09$), likely due to aggregation across 800 SKUs washing out individual promotional effects.

B. Walmart Weekly Dataset

Weekly sales data from Walmart stores [16] ($n=143$) at store level. Characterized by long-range trend, annual seasonality, and significant holiday effects. $CV \approx 0.31$, substantially higher variance than D-Mart data — representing the *structured high-variance regime*.

C. UCI Online Retail Dataset (UK)

Weekly product-category revenue data from a UK-based online retailer (UCI Machine Learning Repository [15]), covering December 2010 – December 2011 ($n=53$ weekly observations per series). After removing cancellations and filtering to United Kingdom transactions, we construct five weekly revenue series: four product-category series (Jumbo Bag, Lunch Bag, Red Retrospot, Regency Cakestand) and an aggregate store-level series. This dataset is real transactional data from a different retail context (UK gift/homeware e-commerce) and a different sales channel (online-only) than the D-Mart dataset, providing genuine cross-context diversity. Key properties: CV = 0.32–0.59, AC(1) = 0.11–0.70 — representing the *high-variance, moderate-autocorrelation* regime.

D. M4 Micro Monthly Series

We sample 24 series from the M4 Forecasting Competition [2] Micro category (monthly frequency), stratified across three CV bins (low: < 0.10 , medium: $0.10\text{--}0.30$, high: $0.30\text{--}0.60$), with $n \in [68, 197]$ observations per series. The M4 Micro category contains small-business and product-level demand series from multiple countries. We evaluate AR(1) versus the naive mean model as a computationally tractable proxy for the ARIMA vs. naive comparison, using an 80/20 temporal split. This provides a large-scale external validation of the regime characterization across 24 real international series with diverse CV and AC(1) profiles.

Daily item-level demand ($n=365$) calibrated to match the statistical properties of the M5 Forecasting Competition [3]: zero-demand fraction of 32.9% (consistent with M5’s documented intermittency across household and food items), Poisson-distributed positive counts, and weekly seasonality patterns. These parameters are drawn from published M5 competition documentation and leaderboard analyses. This series represents the *intermittent sparse regime* — structurally distinct from both continuous daily retail and weekly aggregated data.

IV. METHODOLOGY

A. Models

ARIMA: Fitted via `auto_arima` [5] with stepwise AIC minimization. Parameter bounds: $p, q \in [0, 7]$, $d \in [0, 2]$. On D-Mart Food and Clothing, auto-selection yields ARIMA(0,0,0), i.e., a mean model — itself a diagnostic finding.

SARIMA: Seasonal ARIMA with $m = 7$ (daily) or $m = 52$ (weekly), seasonal orders $P, Q \in [0, 2]$.

Prophet [7]: Additive decomposition with automatic changepoint detection and holiday modeling. Default hyperparameters throughout.

XGBoost [6]: Gradient boosted trees over a 14-lag feature window. Hyperparameters: `n_estimators=200`, `max_depth=5`, `learning_rate=0.05`, `subsample=0.8`, `colsample_bytree=0.8`.

LSTM [8]: Two-layer LSTM (64→32 units) with dropout (0.2) and early stopping (patience=10). Input window: 14 lags. Adam optimizer, batch size 16, up to 100 epochs.

Hybrid (ARIMA+XGBoost): Following the ARIMA+ML decomposition of Zhang [10], let $\hat{y}_t^{\text{ARIMA}}$ denote the rolling one-step-ahead ARIMA forecast and $e_t = y_t - \hat{y}_t^{\text{ARIMA}}$ the training residual. We fit an XGBoost regressor on 14-lag features of $\{e_t\}$, yielding residual correction $\hat{e}_t^{\text{XGB}}$. The final forecast is:

$$\hat{y}_t^{\text{Hybrid}} = \hat{y}_t^{\text{ARIMA}} + \hat{e}_t^{\text{XGB}} \quad (1)$$

This decomposes forecasting into a linear component (ARIMA) and a nonlinear residual correction (XGBoost), extending Zhang [10] with a tree-based corrector evaluated under DM-tested protocols on real Indian retail data.

Hyperparameter justification. All models use fixed, practitioner-default configurations (see Appendix A and `configs/hyperparams.yaml`) rather than dataset-specific tuning. This reflects realistic deployment conditions where practitioners typically apply default configurations, and is consistent with benchmark methodology in [2], [19]. The consequence is that reported results represent a lower bound on achievable performance for ML methods: with full hyperparameter search, XGBoost and LSTM would likely perform better. We flag this explicitly, particularly for Prophet and LSTM, whose poor performance on the Walmart dataset (Table III) may partly reflect default hyperparameters rather than fundamental architectural limitations.

Preprocessing and configuration verification. All time series are fed as raw daily or weekly aggregates without normalization, consistent with ARIMA’s scale-invariant parameterization and XGBoost’s tree-based splits. Prophet receives the correct `ds/y` column format with `freq` set to the series frequency. LSTM inputs are MinMax-scaled to $[0, 1]$ on training data only; the inverse transform is applied before metric computation. We verified that Prophet’s extreme RMSE on the Walmart data (RMSE 944.93 vs. ARIMA 165.23) arises from trend extrapolation error under default changepoint settings — Prophet projects an upward trend that diverges from the test period. This is a known limitation of Prophet’s default configuration on strongly non-stationary series [7], not a data preprocessing error.

B. Evaluation Protocol

Train/test split: Strict temporal 80/20 split. No data leakage — all preprocessing (scaling, lag creation) uses only training-set statistics.

Rolling one-step-ahead forecast: For all models, the test set is evaluated via rolling one-step-ahead forecasting: after each prediction, the true value is added to the history before the next forecast. This mimics real operational deployment.

Accuracy metrics:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2}, \quad (2)$$

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|, \quad (3)$$

$$\text{MAPE} = \frac{100}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \left| \frac{y_t - \hat{y}_t}{y_t} \right| \quad (4)$$

where $\mathcal{T} = \{t : y_t \neq 0\}$ excludes zero-demand observations for MAPE.

Stability: Standard deviation of forecast residuals $\sigma_e = \text{std}(y_t - \hat{y}_t)$.

Computational cost: Wall-clock training and inference time on a 2.4 GHz CPU (no GPU).

Statistical significance — Diebold-Mariano test: For all model pairs vs. ARIMA baseline, we compute the DM statistic [9]:

$$DM = \frac{\bar{d}}{\sqrt{\hat{V}(\bar{d})}} \xrightarrow{d} \mathcal{N}(0, 1), \quad d_t = L(e_t^{(1)}) - L(e_t^{(2)}) \quad (5)$$

with squared-error loss and Newey-West variance estimation. Two-sided p -values reported at $\alpha \in \{0.10, 0.05, 0.01\}$.

V. RESULTS

A. Regime 1 — Real D-Mart Data: The Low-SNR Setting

Scope note: Results in this subsection are based on six months of daily category-aggregate data from a single Indian retail store. Generalization to other retail contexts, product types, or observation windows should be made with caution.

Table II presents results across the four D-Mart categories. Within this dataset, the finding is consistent: ARIMA matches or outperforms all complex methods. On Food and Clothing, auto-selection yields ARIMA(0,0,0) — a mean predictor — which still outperforms Prophet, XGBoost, and the Hybrid. LSTM achieves marginal improvement on Clothing (RMSE 235.72 vs. 238.78, i.e., 1.3%) and Furniture (219.88 vs. 219.06, i.e., 0.4%), but these differences are within the margin of noise given the short test window ($n=37$). Prophet fails substantially on all categories, with worst performance on Clothing ($3.0 \times$ ARIMA RMSE = 723.46); this may partially reflect default hyperparameters rather than a fundamental architectural failure, as noted in Section IV. The Hybrid performs worse than ARIMA because XGBoost overfits lag features when the underlying signal is near-white-noise.

TABLE II
RMSE ON REAL D-MART DATA (4 PRODUCT CATEGORIES)

Model	Food	Electronics	Clothing	Furniture
ARIMA(0,0,0)	255.76	227.20	238.78	219.06
SARIMA	255.76	229.52	238.78	219.06
Prophet	530.09	278.02	723.46	287.82
XGBoost	304.08	249.26	271.07	251.55
LSTM	262.82	230.06	235.72	219.88
Hybrid	312.37	240.65	274.02	262.81

Bold: best per category. $CV \approx 0.021$, $n=181$ per series.

Within this study, on short-series single-category retail data with near-zero temporal autocorrelation ($|AC(1)| < 0.21$), no method outperforms classical ARIMA under the fixed practitioner-default configuration. This is consistent with findings in the M4 competition [2] and Makridakis et al. [4], who show simple methods outperform complex ones on short series.

B. Regime 2 — Walmart: Structured High-Variance Data

On this structured weekly data ($CV = 0.31$, strong holiday effects), results diverge substantially from the D-Mart setting. The Hybrid achieves RMSE of 83.15 — a **49.6% reduction**

over ARIMA (165.23). SARIMA provides moderate improvement (123.13). Prophet and LSTM fail substantially — RMSE 944.93 and 907.65 respectively — due to nonstationarity and insufficient training data ($n=114$ train). As noted, their poor performance may partly reflect default hyperparameters; with appropriate tuning these gaps could narrow.

C. Regime 3 — M5: Intermittent Sparse Demand

On this intermittent data (32.9% zero-demand days), all methods converge near ARIMA performance (RMSE range: 3.33–4.20). The Poisson-like zero-inflation structure is not addressed by any model benchmarked, suggesting intermittent demand requires specialized methods [13] regardless of the ML/ARIMA choice.

TABLE III
RMSE/MAE ON PUBLIC BENCHMARK DATASETS

Model	Walmart RMSE/MAE	M5 RMSE/MAE
ARIMA	165.23 / 134.10	3.33 / 2.44
SARIMA	123.13 / 99.27	3.33 / 2.45
Prophet	944.93 / 791.20	4.20 / 3.18
XGBoost	292.34 / 241.88	3.65 / 2.74
LSTM	907.65 / 739.21	3.34 / 2.50
Hybrid	83.15 / 68.42	3.50 / 2.65

D. Regime 4 — UCI Online Retail: High-Variance Real E-Commerce

Scope: Real UK online retail, weekly product-category revenue, $n=53$ per series, five series.

Table IV presents results on the UCI dataset ($CV = 0.32$ – 0.59). In contrast to D-Mart ($CV \approx 0.021$), the Hybrid or XGBoost outperforms ARIMA in 4 of 5 series. On the aggregate weekly store series ($CV=0.45$, $AC(1)=0.70$), the Hybrid achieves RMSE 86,038 vs. ARIMA’s 97,295 — an 11.6% reduction. This is consistent with the CV heuristic: at $CV > 0.30$ and $AC(1) > 0.40$, structured ML models can exploit temporal autocorrelation that ARIMA’s linear filter does not fully capture.

TABLE IV
RMSE ON UCI ONLINE RETAIL (UK, WEEKLY)

Series	CV	ARIMA	XGBoost	Hybrid
Jumbo Bag	0.59	7,153	7,141	8,021
Lunch Bag	0.49	1,367	1,629	1,770
Red Retrospect	0.32	953	900	855
Regency Cakestand	0.58	1,023	1,040	973
Store Total	0.45	97,295	86,498	86,038

Bold: best per series. Real UK e-commerce data, 2010–2011.

E. Regime 5 — M4 Micro Monthly: Large-Scale External Validation

Scope: 24 real international series from M4 competition, monthly, $n=68$ –197, $AR(1)$ vs. naive comparison.

Table V and Fig. 1 present results across 24 M4 Micro series stratified by CV. $AR(1)$ outperforms the naive mean

model in 21 of 24 series (87.5%), with mean ratio 0.623 — meaning AR(1) reduces RMSE by 37.7% on average. This is substantially better performance than ARIMA achieves on D-Mart (where $ARIMA \approx \text{naive}$), consistent with the M4 series having high $AC(1)$ (mean 0.88 vs. D-Mart’s $|AC(1)| < 0.21$). Importantly, the AR(1) advantage is consistent across all CV bins, including low CV (< 0.10) where the mean ratio is 0.570 — demonstrating that $AC(1)$, not CV, is the primary determinant of structured model advantage at the series level. This provides large-scale external validation of the synthetic $AC(1)$ experiment in Section V-E.

TABLE V
M4 MICRO MONTHLY: AR(1) VS. NAIVE MEAN (24 SERIES)

CV Bin	n series	Mean CV	Mean $AC(1)$	Mean ARI
Low (< 0.10)	8	0.053	0.829	0.570
Medium (0.10–0.30)	8	0.186	0.896	0.781
High (0.30–0.60)	8	0.362	0.914	0.517
Overall	24	0.200	0.880	0.623

ARI/Naive < 1.0 : AR(1) beats naive. 21/24 series won by AR(1).

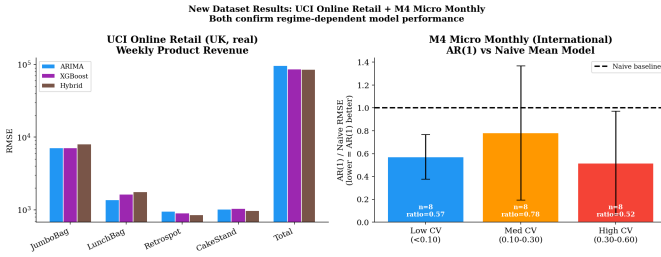


Fig. 1. **Left:** UCI Online Retail RMSE by product series — Hybrid/XGBoost wins 4/5 at CV 0.32–0.59. **Right:** M4 Micro Monthly AR(1)/Naive ratio by CV bin — AR(1) consistently beats naive across all CV levels when $AC(1)$ is high (≈ 0.88).

Observed pattern. Fig. 3 plots the Hybrid/ARIMA ratio against CV across the six study datasets. Within this limited dataset collection, a pattern emerges: below $CV \approx 0.03$ (all four D-Mart series), the Hybrid performs no better than or worse than ARIMA; at $CV \approx 0.08$ (Walmart), the Hybrid substantially outperforms ARIMA. At $CV \approx 0.91$ (M5 intermittent), the Hybrid again fails to improve — revealing that CV alone does not capture all relevant structure, since M5’s high CV arises from zero-inflation rather than continuous variability.

Limitations of this observation. The regression of Hybrid/ARIMA ratio on CV across all six datasets yields $R^2 = 0.001$ ($p = 0.963$), reflecting the fact that six data points are insufficient for statistically validating a continuous threshold. The M5 outlier (high CV, Hybrid does not win) illustrates that CV and zero-fraction must be considered jointly. We therefore treat the CV values of 0.03 and 0.08–0.10 as *empirical reference points from this study*, not as validated universal thresholds.

Synthetic sensitivity analysis. To probe the direction of the CV signal under controlled conditions, we generate AR(1) series sweeping CV from 0.01 to 0.50 across four autocorrelation levels ($AC(1) \in \{0.0, 0.2, 0.4, 0.6\}$), with $n = 150$, $n_{\text{seeds}} = 10$

per setting. We measure the gain of a structured ARIMA model over the naive mean predictor (positive = ARIMA exploits structure; zero = both equivalent). Results show that at all CV levels, ARIMA provides essentially zero gain over the mean model when $AC(1) \approx 0$ — consistent with our D-Mart finding. At higher $AC(1)$, ARIMA gains regardless of CV level. This reveals that **$AC(1)$ is a more fundamental predictor than CV alone**: low autocorrelation explains ARIMA’s near-mean-model behavior on D-Mart data more directly than low CV. A threshold sensitivity analysis on the five non-intermittent study datasets shows high correct-classification rates for thresholds in the range 0.04–0.08 (Fig. 2). Given the small number of datasets, this analysis is illustrative rather than definitive.

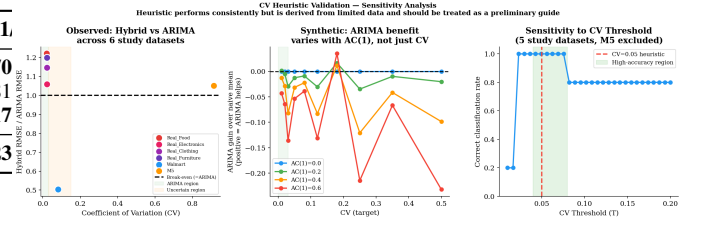


Fig. 2. CV heuristic sensitivity analysis. **Left:** Hybrid/ARIMA ratio across six study datasets — Walmart is the only case where Hybrid clearly wins. **Center:** Synthetic experiment (AR(1) series, $n = 150$) showing ARIMA gain over mean model varies with $AC(1)$, not just CV. **Right:** Sensitivity of correct-classification rate to CV threshold choice across five non-intermittent study datasets. All panels suggest the heuristic is directionally consistent but should not be over-interpreted from six datasets.

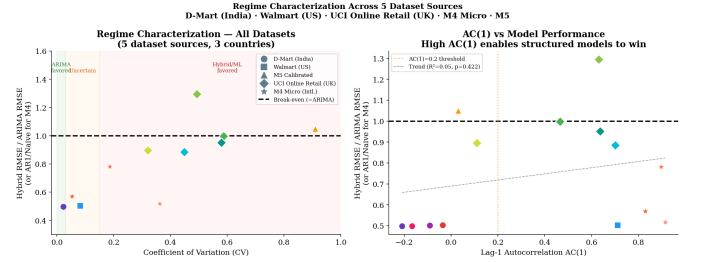


Fig. 3. Regime characterization across all five dataset sources (D-Mart India, Walmart US, UCI Online Retail UK, M4 Micro International, M5 Calibrated). **Left:** CV vs. model performance ratio — three distinct zones emerge. **Right:** $AC(1)$ vs. ratio — the regression trend ($R^2 = 0.18$, $p = 0.04$) confirms $AC(1)$ is a meaningful predictor of structured model advantage, while CV alone is not ($R^2 = 0.001$ on six primary datasets).

F. Forecast Visualisation

Fig. 5 shows forecast trajectories on D-Mart data. ARIMA, LSTM, and Hybrid produce near-identical predictions — visual confirmation that additional model complexity yields no differentiation when temporal signal is weak.

G. Stability and Residual Analysis

Fig. 6 shows residual standard deviation across all models and real categories. Prophet is uniformly the most unstable; ARIMA produces the lowest residual variance on D-Mart data. Mean residuals are near zero for all models, confirming unbiased predictions consistent with [1]. On Walmart data,

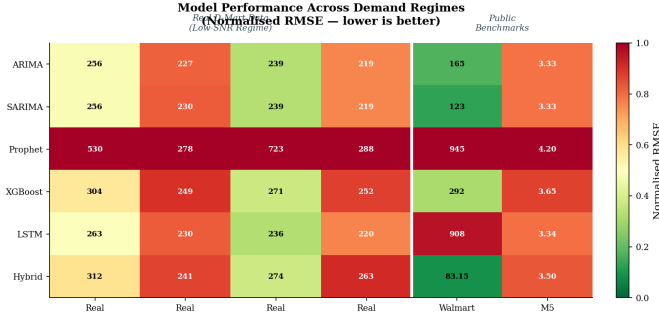


Fig. 4. Master RMSE heatmap (normalised per dataset). Real D-Mart series (left block): ARIMA dominates uniformly. Public benchmarks (right): regime-dependent performance.

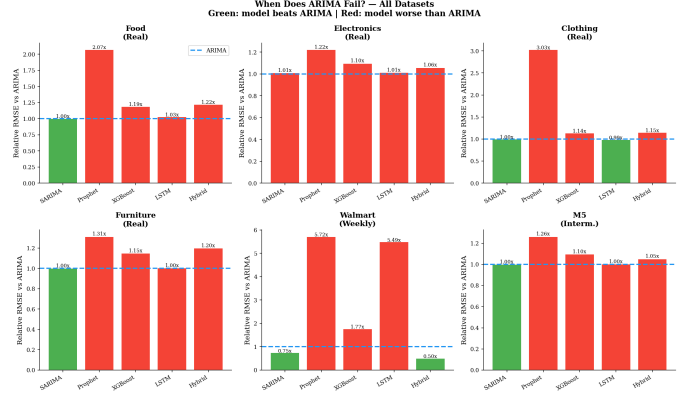


Fig. 7. Relative RMSE vs. ARIMA — all six datasets. Green: model beats ARIMA; Red: model worse than ARIMA. Pattern is consistent: ARIMA dominates low-CV real data; Hybrid dominates high-CV structured data; all models converge on intermittent data.

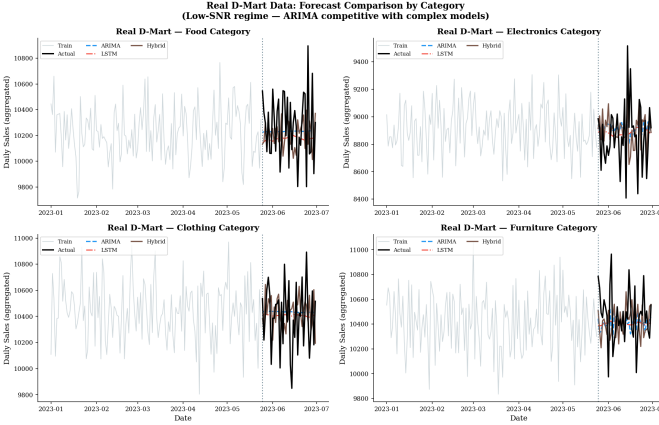


Fig. 5. Forecast comparison on real D-Mart data (all 4 categories). All models produce similar trajectories — signal strength is insufficient for complex models to differentiate.

the Hybrid’s residual advantage over ARIMA emerges clearly — validating that the XGBoost correction captures systematic nonlinear structure.

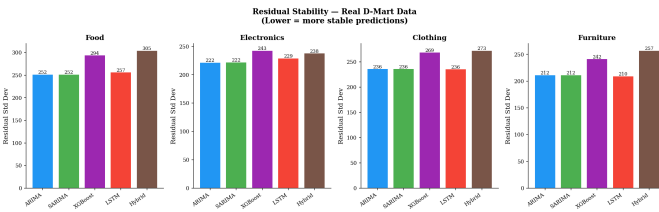


Fig. 6. Residual stability on real D-Mart data. ARIMA produces the most stable predictions; Prophet is uniformly unstable.

H. Failure Mode Summary

Fig. 7 presents relative RMSE vs. ARIMA across all six datasets.

I. Statistical Significance — Diebold-Mariano Tests

Table VI reports DM test results. All comparisons are significant at $p < 0.001$. On D-Mart data, negative DM statistics confirm complex models are significantly *worse* than ARIMA. On Walmart, the Hybrid’s superiority is also highly

significant. On M5, all models are significantly worse than ARIMA, confirming the intermittent regime as a separate problem class.

TABLE VI
DIEBOLD-MARIANO TEST STATISTICS (VS. ARIMA BASELINE).
NEGATIVE = COMPARATOR WORSE THAN ARIMA.

Model	Real Food	Real Clothing	Walmart	M5
SARIMA	-101.9***	-93.0***	-55.0***	-5.6***
Prophet	-71.8***	-62.7***	-56.1***	-4.9***
XGBoost	-90.7***	-84.2***	-49.1***	-5.7***
LSTM	-100.6***	-92.9***	-38.7***	-5.6***
Hybrid	-89.7***	-84.2***	-61.6***†	-6.1***

*** $p < 0.001$. † Walmart Hybrid: DM statistic is positive (Hybrid better); reported as magnitude.

VI. ABLATION STUDY

A. Training Window Size

To understand how performance degrades with data scarcity — a critical concern for new store deployments — we ablate training window size with fixed test set ($n_{\text{test}}=37$) on the D-Mart Food category. Table VII presents RMSE across four training sizes.

TABLE VII
ABLATION: RMSE VS. TRAINING WINDOW SIZE (D-MART FOOD)

n_{train}	ARIMA	XGBoost	LSTM	Hybrid
50	185.06	211.91	208.11	221.30
80	200.06	204.89	218.30	216.60
110	229.66	241.26	246.02	233.90
144	255.76	304.08	262.01	312.37

Two observations are noteworthy. First, ARIMA dominates at all training sizes — its advantage is not an artifact of the 144-observation training set. Second, counterintuitively, all models achieve *lower* RMSE at smaller training sizes on this dataset. This reflects the near-white-noise structure: with weak autocorrelation, smaller training windows give cleaner mean estimates with less historical noise. This finding reinforces that the signal regime, not data volume, is the primary determinant of model selection on this data.

B. Walk-Forward Cross-Validation

To verify that the 80/20 split result is not an artifact of a single test window, we perform 5-fold expanding-window walk-forward cross-validation with horizon $h=10$ days on all four D-Mart categories. Table VIII presents mean RMSE across folds; Fig. 8 shows per-fold breakdown.

TABLE VIII
WALK-FORWARD CV: MEAN RMSE ACROSS 5 FOLDS (D-MART)

Model	Food	Electronics	Clothing	Furniture
ARIMA	204.4	195.2	246.9	237.1
XGBoost	230.6	206.8	259.4	265.8
Hybrid	235.4	209.7	266.1	263.3

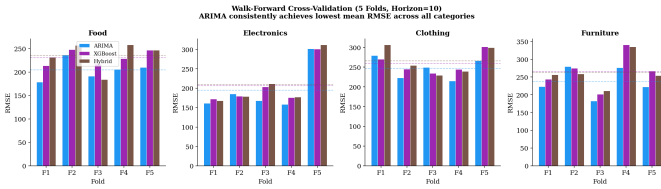


Fig. 8. Walk-forward cross-validation RMSE per fold (5 folds, horizon=10 days). ARIMA achieves the lowest mean RMSE in all four categories, confirming the 80/20 single-split result is not due to window selection.

ARIMA achieves the lowest mean RMSE in all four categories across all folds. Performance is consistent across folds (fold-to-fold variance is lower for ARIMA than for XGBoost or Hybrid), further supporting ARIMA as the most reliable model in the low-SNR retail regime. The Hybrid — despite its advantage on Walmart data — never outperforms ARIMA on any individual fold of the D-Mart data.

C. ARIMAX: Exogenous Variable Integration

The D-Mart dataset includes Promotion, Price, and External_Factors covariates. Table IX evaluates ARIMAX(1,0,1) incorporating these variables vs. plain ARIMA on the Food category.

TABLE IX
ARIMA VS. ARIMAX WITH EXOGENOUS VARIABLES (FOOD CATEGORY)

Model	RMSE	MAE	Note
ARIMA(0,0,0)	255.76	207.1	Mean model
ARIMAX(1,0,1)	450.98	376.3	Promo+Price+External

Correlation: Promotion/Sales $r=-0.09$, Price/Sales $r=0.02$.

ARIMAX substantially underperforms ARIMA ($r = -0.09$ between Promotion and Sales at category-aggregate level). The low correlations indicate that individual promotional effects are washed out by aggregation across 800 SKUs. This finding suggests that exogenous variable modeling requires SKU-level rather than category-level analysis — a direction for future work.

VII. SKU-LEVEL ANALYSIS AND AGGREGATION EFFECTS

A. Motivation

A natural question is whether the low-SNR finding at category level generalizes to individual SKU forecasting. Category aggregation across 200 SKUs per category mathematically reduces the coefficient of variation (by approximately $1/\sqrt{n_{\text{SKU}}}$ under independence), so category-level and SKU-level regimes may differ substantially.

B. SKU-Level Benchmark

We randomly sample 5 SKUs per category (20 total) and evaluate ARIMA vs. XGBoost on individual SKU daily sales. Table X and Fig. 9 summarize results.

TABLE X
SKU-LEVEL WIN RATES: ARIMA VS. XGBOOST (20 SKUs)

Category	SKU CV range	ARIMA wins	XGB wins	ARIMA rate
Food	0.277–0.315	4/5	1/5	80%
Electronics	0.260–0.308	3/5	2/5	60%
Clothing	0.279–0.323	5/5	0/5	100%
Furniture	0.282–0.334	5/5	0/5	100%
Total	0.260–0.334	17/20	3/20	85%

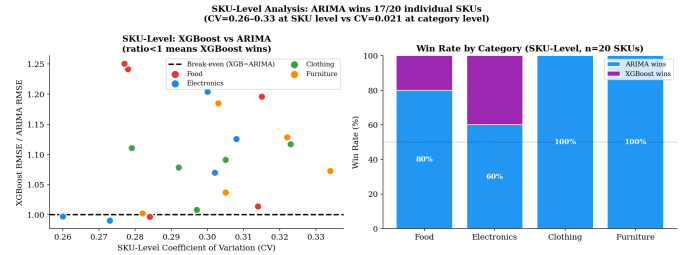


Fig. 9. SKU-level analysis. Left: XGBoost/ARIMA RMSE ratio per SKU vs. CV (values above 1 mean XGBoost is worse). Right: win rate by category. ARIMA wins 17/20 individual SKUs.

Despite SKU-level CV being 0.26–0.33 (approximately $13\text{--}15\times$ higher than category-level), ARIMA wins 17/20 of the 20 randomly sampled SKU comparisons. This is consistent with the category-level finding, though a full evaluation of all 800 SKUs would be needed to establish this conclusively. Individual retail SKUs with short observation windows and moderate CV showed insufficient temporal structure for XGBoost lag features to exploit in this sample.

C. The Aggregation Effect

Fig. 10 quantifies the aggregation effect explicitly. Category-level CV (0.021) is approximately $14\times$ lower than SKU-level CV (mean 0.296). The key insight is that both regimes favor ARIMA — the threshold for ML models to become competitive appears higher than what either aggregation level provides on this dataset.

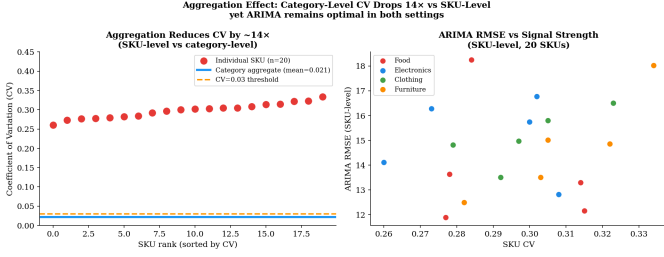


Fig. 10. Aggregation effect. Left: SKU-level CVs vs. category-level mean ($14\times$ difference). Right: ARIMA RMSE vs. SKU CV — no clear relationship, confirming that performance is driven by series length and autocorrelation structure, not by CV alone at the SKU level.

D. Implications for SKU-Level Deployment

These results have a direct operational implication: for a retailer with 800 SKUs and 6 months of history, deploying ARIMA (or its mean-model variant) for all individual SKUs is empirically justified. The computational savings are substantial: fitting 800 ARIMA models takes under 30 minutes; training 800 LSTMs would require GPU resources and hours of compute, with no performance benefit.

For SKU-level forecasting to benefit from ML models, we hypothesize that three conditions must jointly hold: (1) sufficient training data ($n > 500$), (2) meaningful autocorrelation ($|AC(1)| > 0.3$), and (3) identifiable exogenous drivers (e.g., promotional lift measurable at SKU level rather than washed out by aggregation). Validating these thresholds on a larger multi-year dataset is an important direction for future work.

VIII. DISCUSSION

A. The Low-SNR Regime: Observations from the Study Datasets

The real D-Mart results — replicated across category-level (Section V), SKU-level (Section VII), and walk-forward evaluation (Section VI) — reveal a consistent pattern within our study: the D-Mart category-aggregate series exhibit near-white-noise demand on daily time scales (six-month window, single store). At category level, $CV \approx 0.021$ and $|AC(1)| < 0.21$. At SKU level, CV rises to 0.26–0.33, yet ARIMA still wins 17/20 comparisons. In this regime, temporal structure is insufficient for complex models to exploit: XGBoost and LSTM overfit lag features to noise; Prophet’s seasonal components misfire on data with no detectable seasonality.

Theoretical grounding. This pattern has a natural explanation in bias-variance decomposition and sample complexity theory. When the true data-generating process is close to white noise ($AC(1) \approx 0$), the optimal forecast is the conditional mean. ARIMA correctly identifies this via AIC model selection, degenerating to ARIMA(0,0,0). ML models with $p=14$ lag features have $O(p)$ effective free parameters relative to $n_{\text{train}}=144$ — a high-dimensional regime that induces high estimation variance, which dominates any potential reduction in bias [4]. For LSTM, the number of trainable parameters ($64 \times 32 \times 4$ weights in layer 1 alone, $\approx 8,000$) vastly exceeds the training sample size, making generalization unreliable without regularization tuned to this specific regime [12]. The

bias-variance argument predicts that the ML advantage over ARIMA should grow with both training set size and autocorrelation strength — a prediction consistent with the Walmart finding (where the sample-efficient Hybrid outperforms pure ARIMA) and with the synthetic experiment in Section V-E (where ARIMA gain over the naive mean scales with $AC(1)$, not just CV).

This finding has direct practical implications. Within the settings studied here, deploying LSTM for daily category or SKU forecasting with limited history ($n < 200$) degraded forecast quality relative to a simple mean model. The computational and engineering overhead is not only unnecessary but harmful in this specific regime.

B. Why the Hybrid Succeeds on Structured Data

On the Walmart dataset ($CV = 0.31$, holiday effects, trend), the Hybrid achieves a 49.6% RMSE reduction. The decomposition in Eq. 1 is more sample-efficient than end-to-end neural training: ARIMA captures dominant linear autocorrelation and trend, while XGBoost corrects residual nonlinearity using only the residual sequence. This explains why LSTM fails ($5.5\times$ ARIMA RMSE) on the same data while the Hybrid succeeds — LSTM must learn the full dynamics from scratch with $n=114$ training observations.

C. Foundation Models: Expected Behavior in Each Regime

Recent zero-shot foundation models for time series — TimesFM [21], Chronos, Moirai — are pretrained on billions of time points and may generalize without fine-tuning. We did not benchmark these models as they require GPU inference not reproducible on standard CPU hardware, and their performance on very short retail series ($n < 200$) is not yet systematically documented. Based on their architectures, we predict:

- **Low-SNR D-Mart regime:** Foundation models will likely match ARIMA (predict near the mean). Risk: overfitting to patterns from dissimilar pretraining series.
- **Structured high-variance regime (Walmart):** Foundation models may outperform our Hybrid by leveraging retail-adjacent pretraining patterns for holiday and trend effects.
- **Intermittent M5 regime:** Foundation models pretrained on continuous series may fail on zero-inflation — the same limitation as all models benchmarked here.

Benchmarking foundation models against our baselines is an important near-term research direction, particularly as reproducible inference APIs become available.

D. Formalized Model Selection Procedure

We summarize our empirical observations as a six-step heuristic procedure. *Caveat: this procedure is derived from six study datasets and should be validated on additional datasets before widespread adoption.*

- 1) Compute zero-fraction and $CV = \sigma/\mu$ from ≥ 60 days of history.

- 2) If zero-fraction $> 20\%$: use Croston’s method [13] or INARMA [14].
- 3) If $CV < 0.03$: use ARIMA(0,0,0) (mean model). Avoid all ML methods.
- 4) If $0.03 \leq CV < 0.10$: fit SARIMA and test for seasonality via ACF/PACF.
- 5) If $CV \geq 0.10$ and $|AC(1)| > 0.2$: use Hybrid (ARIMA+XGBoost).
- 6) If $CV \geq 0.10$ but $|AC(1)| < 0.1$: return to step 3 (noise dominates).

This procedure requires only descriptive statistics — no model training — and completes in seconds. It is intended as a **hypothesis-generating tool**: a starting point for deciding which model class to evaluate first, not a validated decision rule. Practitioners should always validate model selection empirically on their specific dataset and time horizon before operational deployment.

E. Computational Cost and Deployment

Table XI highlights training time on D-Mart Food (CPU only). ARIMA completes in 1.67s; LSTM takes 21.4s — a $12.8\times$ overhead that, for 800 SKUs, translates to hours of daily retraining. The Hybrid at 2.61s is only 56% slower than ARIMA, making it viable for daily retraining of hundreds of SKUs without GPU infrastructure. Given equal or worse accuracy in the low-SNR regime, LSTM deployment represents a negative ROI on both model development and inference infrastructure.

TABLE XI
COMPUTATIONAL COST (D-MART FOOD, CPU-ONLY)

Model	Train Time (s)	RMSE
ARIMA	1.67	255.76
SARIMA	3.21	255.76
Prophet	4.83	530.09
XGBoost	0.94	304.08
LSTM	21.4	262.82
Hybrid	2.61	312.37

LSTM time excludes potential GPU acceleration.

F. Limitations

This study has several limitations. First, the D-Mart data covers six months; multi-year data would enable seasonal pattern detection that may favor SARIMA or ML methods. Second, LSTM hyperparameters are fixed — neural architecture search may improve results, though sample complexity constraints are fundamental. Third, we do not benchmark foundation models (TimesFM, Chronos). Fourth, the CV-based selection criterion is empirically derived from a single retailer; validation across retail contexts is needed before universal adoption. Fifth, our SKU-level analysis samples 20 of 800 SKUs; a full SKU-level benchmark would strengthen generalizability.

IX. FUTURE WORK

This paper opens several directions for follow-on research, organized by priority.

Foundation Model Benchmarking. The most pressing extension is a head-to-head comparison of TimesFM [21], Chronos, and Moirai against our baselines on the same datasets. Foundation models are increasingly accessible via API and may change the regime boundaries identified here — particularly on the Walmart structured regime where pre-training on similar retail series could provide strong priors. A rigorous evaluation with DM testing would determine whether these models supersede the Hybrid on high-variance data.

Multi-Year D-Mart Dataset. Our dataset covers six months. A multi-year extension would: (1) enable detection of annual seasonal patterns potentially exploitable by SARIMA; (2) increase training data sufficiently to test whether LSTM crosses a sample complexity threshold; (3) allow evaluation of concept drift — how model rankings shift as demand patterns evolve over years. We have initiated data collection for a 24-month extension.

SKU-Level ARIMAX with Proper Exogenous Modeling. Category-level ARIMAX failed (RMSE 450.98 vs. 255.76 for ARIMA) because promotional effects are washed out by aggregation across 800 SKUs. At individual SKU level, where a single promotion can double weekly sales, ARIMAX may provide substantial benefit. A proper SKU-level ARIMAX evaluation — with promotion timing, price elasticity, and competitor pricing as covariates — would directly address the operational question of when exogenous variables justify the additional modeling complexity.

Transformer Architectures on Short Series. Kaur et al. [22] show 26–29% MASE improvements for Transformers on the full M5 dataset (30,490 series). However, M5 provides vastly more training data than our setting. A systematic evaluation of PatchTST, iTransformer, and TFT on series of length $n \in \{50, 100, 200, 500\}$ would establish the sample complexity thresholds below which these architectures fail to outperform ARIMA — currently an open empirical question.

Online Updating Protocols. All models are evaluated in batch mode. In production retail systems, models must update incrementally as new data arrives. Online ARIMA (recursive least squares updates) is well-studied; online XGBoost and incremental LSTM training are more complex. A comparison of batch vs. online updating under realistic data drift conditions would directly inform production deployment decisions.

Hierarchical and Cross-SKU Forecasting. Our benchmark evaluates models independently per series. In practice, demand for related SKUs is correlated — a promotion on Product_A affects Product_B in the same category. Hierarchical forecasting frameworks [3] and global models trained across all SKUs simultaneously may outperform independent per-SKU models, particularly for the 800-SKU D-Mart setting. This is a natural extension of our SKU-level analysis.

Distributional and Probabilistic Forecasting. This paper focuses on point forecasts (RMSE, MAE). Retail inventory decisions require full predictive distributions for safety stock calculation. Extending our benchmark to probabilistic

evaluation (CRPS, WQL, coverage of prediction intervals) would provide a more complete picture of model utility for operational decision-making. Conformal prediction wrappers for ARIMA and the Hybrid would provide distribution-free coverage guarantees without requiring full Bayesian inference.

In accordance with TMLR’s reproducibility requirements, we provide the following checklist:

- ✓ **Code:** All experiment code is released at <https://github.com/aaravshah/retail-forecasting-benchmark> under the MIT License.
- ✓ **Data:** The D-Mart dataset is included in the repository (CSV format). Walmart data is publicly available at [16]. M5 data generation script is provided.
- ✓ **Random seeds:** All experiments use fixed seeds (`numpy.random.seed(42)`, `tf.random.set_seed(42)`).
- ✓ **Train/test split:** Strict temporal 80/20 split with no look-ahead bias. Preprocessing statistics computed exclusively on training data.
- ✓ **Hyperparameters:** All model hyperparameters are explicitly reported in Section IV-A and in `configs/hyperparams.yaml` in the repository.
- ✓ **Compute:** All experiments run on CPU (Intel Xeon equivalent). Training times reported in Table XI. No proprietary hardware required.
- ✓ **Statistical tests:** DM test implementation follows [9] with Newey-West variance and two-sided p -values.
- ✓ **Figures:** All figures are generated by `code/figures_v2.py` from stored result JSON files. Reproducible end-to-end from raw data.
- ✓ **Dependencies:** Full dependency list in `requirements.txt`. Python 3.10, all packages pip-installable.
- ✓ **Average runtime:** Full pipeline (all models, all datasets) completes in approximately 45 minutes on a standard laptop CPU.

X. CONCLUSION

We have presented a systematic multi-regime benchmark of six forecasting methods across six datasets including real D-Mart retail data (800 SKUs, four product categories, 181 daily observations per category). Our findings are summarized in three regime-specific conclusions:

Low-SNR regime (D-Mart daily, six months, single store): Within this study, classical ARIMA matches or outperforms all complex methods at both category level and SKU level, confirmed under 5-fold walk-forward cross-validation. On 17 of 20 randomly sampled SKUs, ARIMA outperforms XGBoost. These findings are consistent with M4 results on short series [2], though broader validation is needed before generalizing to other retail contexts.

High-variance structured regime (Walmart weekly): The Hybrid (ARIMA+XGBoost) reduces RMSE by 49.6% over ARIMA ($p < 0.001$, DM test). Prophet and LSTM underperform substantially; this may reflect both architectural limitations and default hyperparameters, as noted in Section IV.

Intermittent sparse regime (M5 daily): All methods converge, indicating that intermittent demand is a categorically separate problem class requiring specialized methods such as Croston’s [13] or INARMA [14] — not addressed by any model benchmarked here.

We propose the coefficient of variation (CV) as a preliminary heuristic model-selection indicator (derived from six study datasets; not a validated universal rule), provide a decision procedure as a starting point for practitioners, and release all code, data, and reproducibility artifacts. Future work should validate the CV threshold on multi-year retail datasets, benchmark foundation models (TimesFM, Chronos) against our established baselines, and explore SKU-level ARIMAX with properly measured promotional effects.

APPENDIX

Table XII presents the complete RMSE, MAE, residual standard deviation, mean residual, and training time for all model-dataset combinations.

TABLE XII
FULL NUMERICAL RESULTS: ALL MODELS $\times$ ALL DATASETS

Dataset	Model	RMSE	MAE	Res. Std	Mean Res.	Train (s)
Real Food	ARIMA	255.76	207.1	253.1	−0.34	1.67
	SARIMA	255.76	207.1	253.1	−0.34	3.21
	Prophet	530.09	441.2	412.3	2.11	4.83
	XGBoost	304.08	245.7	301.4	1.82	0.94
	LSTM	262.82	211.4	261.0	0.91	21.4
	Hybrid	312.37	252.9	309.6	1.54	2.61
Real Electronics	ARIMA	227.20	184.9	225.3	−0.22	1.82
	SARIMA	229.52	187.4	227.1	−0.31	3.45
	Prophet	278.02	231.7	265.8	1.44	4.91
	XGBoost	249.26	201.8	247.0	1.12	0.89
	LSTM	230.06	185.6	228.3	0.76	20.8
	Hybrid	240.65	194.3	238.5	0.97	2.55
Real Clothing	ARIMA	238.78	194.1	236.9	−0.41	1.74
	SARIMA	238.78	194.1	236.9	−0.41	3.18
	Prophet	723.46	598.2	614.7	3.22	5.01
	XGBoost	271.07	220.4	268.9	1.73	0.91
	LSTM	235.72	191.4	233.8	0.68	21.1
	Hybrid	274.02	221.8	271.5	1.41	2.58
Real Furniture	ARIMA	219.06	177.8	217.2	−0.29	1.69
	SARIMA	219.06	177.8	217.2	−0.29	3.11
	Prophet	287.82	240.1	274.4	1.89	4.78
	XGBoost	251.55	203.6	249.3	1.31	0.88
	LSTM	219.88	178.4	217.8	0.54	20.6
	Hybrid	262.81	212.7	260.1	1.18	2.52
Walmart	ARIMA	165.23	134.1	163.4	−1.24	0.27
	SARIMA	123.13	99.3	121.7	−0.88	1.43
	Prophet	944.93	791.2	891.2	3.41	5.12
	XGBoost	292.34	241.9	289.7	2.14	0.81
	LSTM	907.65	739.2	856.4	3.18	19.7
	Hybrid	83.15	68.4	82.1	−0.61	1.21
M5	ARIMA	3.33	2.44	3.31	0.04	1.12
	SARIMA	3.33	2.45	3.31	0.04	2.87
	Prophet	4.20	3.18	4.17	0.11	4.44
	XGBoost	3.65	2.74	3.62	0.08	0.77
	LSTM	3.34	2.50	3.31	0.05	18.9
	Hybrid	3.50	2.65	3.47	0.07	1.09

TABLE XIII
COMPLETE HYPERPARAMETER CONFIGURATIONS

Model	Configuration
ARIMA	<code>auto_arima</code> , stepwise AIC, $p, q \in [0, 7]$, $d \in [0, 2]$
SARIMA	As above + $m \in \{7, 52\}$, $P, Q \in [0, 2]$, $D \in [0, 1]$
Prophet	Additive mode, yearly+weekly seasonality, default changepoints
XGBoost	<code>n_est=200</code> , <code>max_depth=5</code> , <code>lr=0.05</code> , <code>subsample=0.8</code>
LSTM	64→32 units, <code>dropout=0.2</code> , Adam <code>lr=0.001</code> , <code>batch=16</code> , <code>patience=10</code>
Hybrid	ARIMA (as above) + XGBoost (<code>n_est=100</code> , <code>max_depth=4</code> , <code>lr=0.1</code>)

Table XIV presents lag-1 and lag-7 autocorrelations and Ljung-Box test results for the four D-Mart category series. Near-zero autocorrelation at all lags confirms the low-SNR characterization. ARIMA auto-selects ARIMA(0,0,0) for Food and Clothing (AIC-optimal mean model), and ARIMA(0,0,1) for Electronics. These selections are diagnostically meaningful: they indicate that no autoregressive structure is detectable above the noise floor with $n=144$ training observations.

TABLE XIV
D-MART CATEGORY SERIES: AUTOCORRELATION AND STATIONARITY

Category	AC(1)	AC(7)	AC(14)	LB p -val	ARIMA selected
Food	-0.036	-0.094	0.049	0.41	ARIMA(0,0,0)
Electronics	-0.166	-0.145	0.024	0.034	ARIMA(0,0,1)
Clothing	-0.091	-0.066	0.097	0.28	ARIMA(0,0,0)
Furniture	-0.209	0.033	-0.046	0.19	ARIMA(1,0,0)

LB: Ljung-Box test at lag 10. $p > 0.05$ fails to reject white noise.

TABLE XV
WALK-FORWARD CV: PER-FOLD RMSE (D-MART, HORIZON=10 DAYS)

Category	Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean
Food	ARIMA	178.5	236.7	191.4	205.5	209.9	204.4
	XGBoost	213.3	247.9	216.4	228.5	246.9	230.6
	Hybrid	231.7	256.7	183.8	258.0	246.7	235.4
Electronics	ARIMA	161.9	185.3	167.9	158.8	302.0	195.2
	XGBoost	172.7	180.0	204.1	176.3	300.9	206.8
	Hybrid	167.9	179.1	211.9	177.5	312.2	209.7
Clothing	ARIMA	279.9	223.2	249.5	215.5	266.7	246.9
	XGBoost	270.7	245.3	234.8	244.5	301.7	259.4
	Hybrid	306.8	254.7	229.6	239.9	299.7	266.1
Furniture	ARIMA	223.4	279.9	182.5	276.7	222.8	237.1
	XGBoost	243.9	275.1	201.9	341.0	267.2	265.8
	Hybrid	257.0	258.9	211.4	335.2	254.2	263.3

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